

Sample size & power calculation

A quick visual reference for planning a statistically sound study

Sample size (n)

The number of observations needed to detect a real effect, if one truly exists.

Power (1 - beta)

The probability of correctly detecting an effect that is truly there. Convention: 80%.

The four inputs that determine sample size



Effect size

Bigger effect, fewer subjects needed



Significance (alpha)

Usually 0.05, the false-positive limit



Power

Usually 0.80, chance of detecting it



Variability (sigma)

Noisier data needs more subjects

Classic two-sample formula (continuous outcome)

$$n = 2(z_{\alpha/2} + z_{\beta})^2 \times \sigma^2 / d^2$$

d = minimum detectable difference between groups, per group

Type I and type II errors

H0 true (no real effect)

H0 false (real effect)

Test: no effect

Correct

True negative
probability 1 - alpha

Type II error

Missed real effect
probability beta

Test: effect

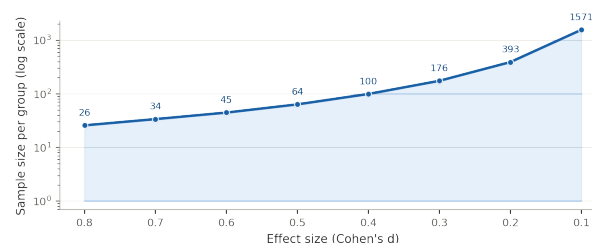
Type I error

False positive
probability alpha

Correct

True positive
probability 1 - beta = power

How sample size grows as the effect gets smaller (alpha=0.05, power=80%)



Quick reference: keep alpha and power fixed, watch how few subjects a large effect needs versus a small one.